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المملكة العربية السعودية

وزارة التعليم

جامعة الباحة

كلية الحاسبات والمعلومات

قسم علوم الحاسبات

Little Quran Stars

Project Report

Mobile Application Development

AL Baha University

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1. Application Overview

Little Quran Stars is an Android application designed to help children memorize the Holy Quran through an engaging and interactive experience. The app allows users to select a specific range of verses from any Surah, configure the number of repetitions per verse, and listen to audio recitation while viewing the corresponding Mushaf page.

Project Phases Implemented:

- Phase 1: UI Design — 8 fully designed screens with consistent green theme and RTL support.
- Phase 2: Events & Navigation — Button-based navigation with click event handlers connecting all screens.
- Phase 3: SQLite & API Integration — local progress storage + AlQuran Cloud API.
- Extra Feature (Group 13): Quiz Mode after each memorization session.

2. Key Features:

- Select any Surah from all 114 Surahs of the Holy Quran.
- Configure verse range (from verse X to verse Y).
- Set custom repetition count per verse.
- Audio playback with 7 different reciters.
- Real-time Mushaf page display.
- Progress saving and resume functionality.
- Night mode and font size settings.
- Quiz mode after memorization sessions.
- How to use guide with swipeable slides.

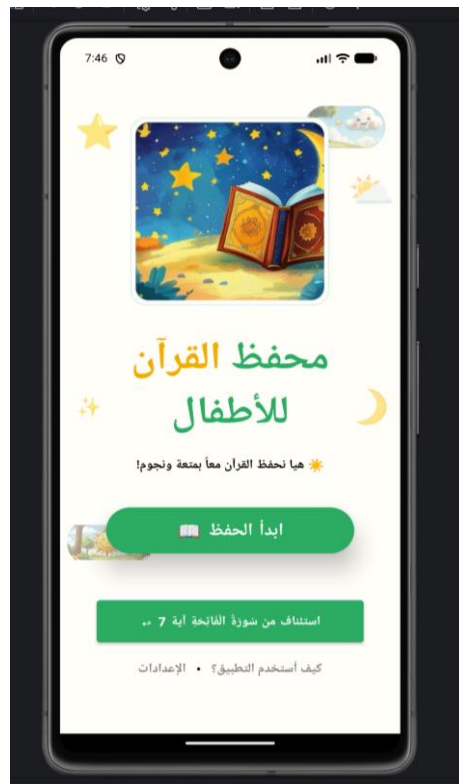
3. Main Screens & User Interface

The application consists of 8 main screens, each designed with a child-friendly interface using a consistent green color palette, clear typography, and intuitive controls. All screens fully support Right-to-Left (RTL) layout for Arabic content, and adapt to both light and dark themes based on the user's preference.

Navigation between screens is handled using Android Intents, with data passed via extras (Surah number, verse range, repeat count).

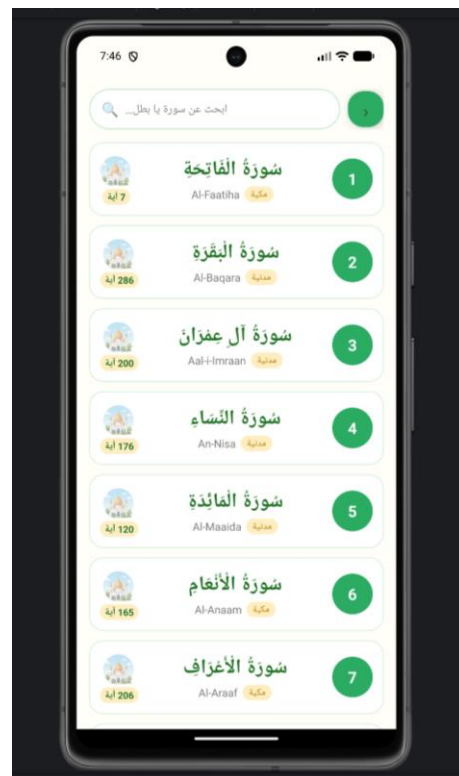
The following sections describe each screen, its purpose, and the main components it contains:

Main Screen (MainActivity):



The home screen displays the app logo, a Start Memorization button, and a Resume button that appears when the user has a saved progress. It also includes links to Settings and How to Use Guide.

Surah List Screen (SurahListActivity):



Displays all 114 Surahs with their names in Arabic and English. Features a search bar to quickly find any Surah. Tapping a Surah navigates to the memorization settings screen.

Memorization Settings Screen (MemorizationSettingsActivity):



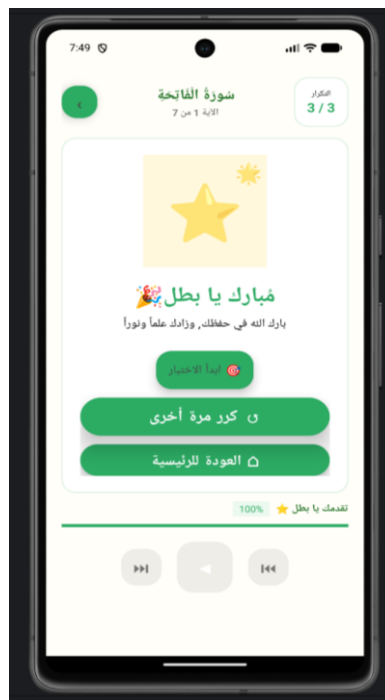
Allows the user to configure: start verse number, end verse number (with +/- buttons), and repeat count per verse (using a NumberPicker). Includes a Start Now button to begin memorization.

Memorization Screen (MemorizationActivity):



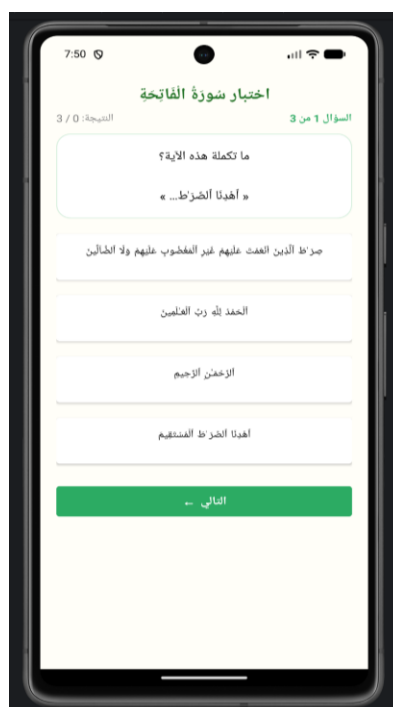
The core screen. Displays: current Surah name and verse progress, repeat counter, Mushaf page image, verse text (fetched from AlQuran Cloud API), progress bar, playback controls (play/pause, next, previous), and a Pause & Resume Later button.

Completion Screen (CompletionActivity):



Shown after finishing all verses in the selected range. Displays a congratulatory message with a star animation, the verse range, repeat count, and buttons to Start Quiz, Repeat Again, or Return to Home.

Quiz Screen (QuizActivity):



The extra feature screen. Shows 3 multiple-choice questions about the memorized verses. Each question displays the beginning of a verse and 4 answer options. Correct answers turn green, wrong answers turn red. Final score is shown in a dialog.

Settings Screen (SettingsActivity):



Allows configuration of: night mode toggle, repeat delay (2/4/6 seconds), font size (small/medium/large), and reciter selection from 7 available reciters.

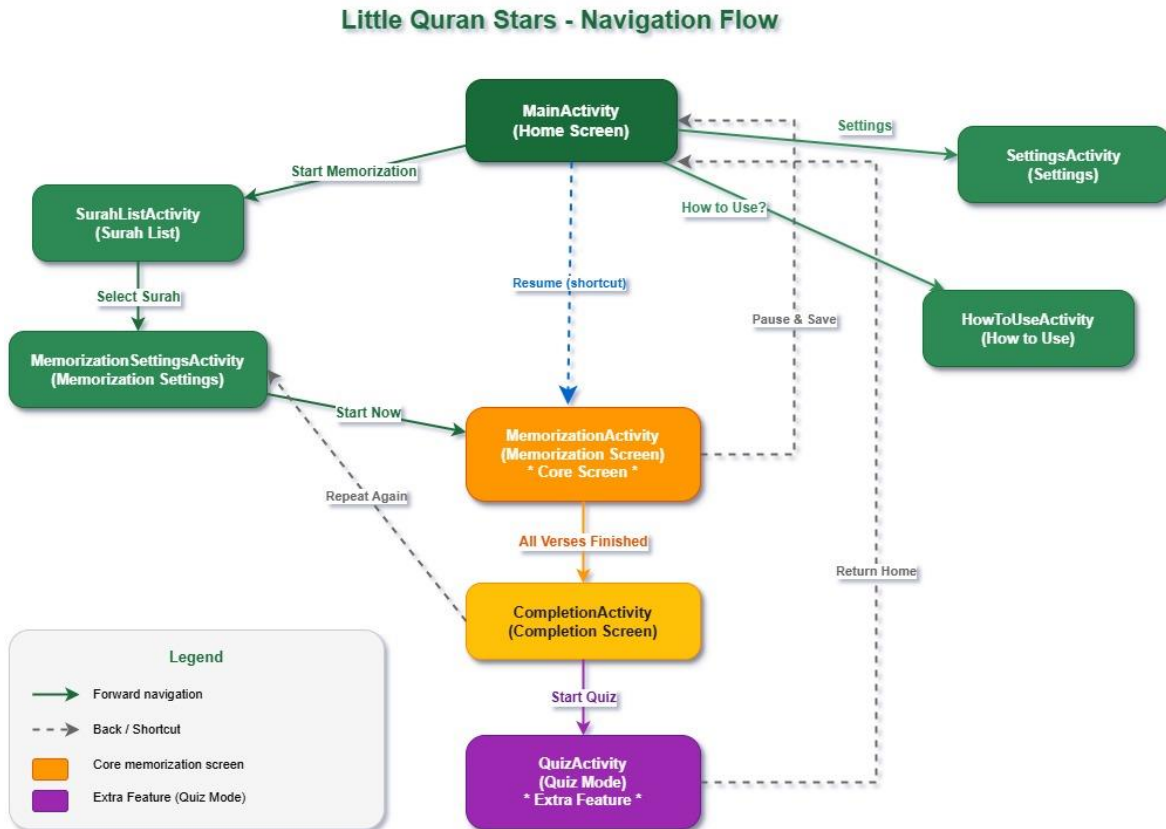
How to Use Screen (HowToUseActivity):



A ViewPager2-based onboarding screen with 4 swipeable slides explaining how to use the app, with emoji illustrations and dot indicators.

4. Navigation Flow

The application follows a clear navigation hierarchy starting from MainActivity. Users can either start a new memorization session or resume their previous progress. The diagram below illustrates all transitions between activities.



Navigation Logic:

- MainActivity → SurahListActivity (via "Start Memorization" button)
- SurahListActivity → MemorizationSettingsActivity (via Surah selection)
- MemorizationSettingsActivity → MemorizationActivity (via "Start Now" button)
- MemorizationActivity → CompletionActivity (automatically after the last verse)
- CompletionActivity → QuizActivity (via "Start Quiz" button)
- MainActivity → MemorizationActivity (shortcut via "Resume" button using saved progress from SQLite)
- MemorizationActivity → MainActivity (via "Pause & Resume Later" - saves state and exits)
- MainActivity → SettingsActivity / HowToUseActivity (via button links)

All transitions use Android Intents with extras to pass data (Surah number, verse range, repeat count) between activities.

5. Database Structure (SQLite)

The app uses SQLite (version 3) as local storage via the **DatabaseHelper** class. The database is named **QuranApp.db** and contains two tables:

5.1 Verses Table:

Stores verse text and page data fetched from the API for offline access.

Column	Type	Description
id	INTEGER PK	Auto-increment primary key
surah_number	INTEGER	Surah number (1-114)
verse_number	INTEGER	Verse number within the Surah
page_number	INTEGER	Mushaf page number
verse_text	TEXT	Arabic text of the verse

5.2 User Progress Table:

Column	Type	Description
id	INTEGER PK	Auto-increment primary key
surah_number	INTEGER UNIQUE	Surah number (unique per Surah)
current_verse	INTEGER	Last verse the user reached
start_verse	INTEGER	Start of the selected range
end_verse	INTEGER	End of the selected range
repeat_count	INTEGER	Configured repetitions per verse
completed_repeats	INTEGER	Repetitions completed at last save
surah_name_ar	TEXT	Arabic name of the Surah
last_updated	DATETIME	Timestamp of last update

Stores each Surah's memorization progress independently, allowing multiple Surahs to have saved positions.

5.3 Key Database Functions:

saveProgress() — Saves or updates the user's position for a specific Surah using INSERT OR REPLACE (CONFLICT_REPLACE).

getLastOpenedSurah() — Returns the most recently accessed Surah ordered by last_updated DESC, used for the Resume button.

addVerse() — Inserts a verse fetched from the API into the verses table for local storage.

getVerse() — Retrieves a specific verse by Surah and verse number from local storage.

6. AlQuran Cloud API Integration

The app integrates with the AlQuran Cloud API (api.alquran.cloud) as the primary external data source for Quran content.

All API calls are made asynchronously on background threads to prevent blocking the UI.

Base URL: <https://api.alquran.cloud/v1/>

6.1 Endpoints Used

The list of all 114 Surahs is stored locally in the SurahModel class. The API is used only for fetching verse text, audio URLs, and Mushaf page numbers as detailed below:

Endpoint	Purpose	Example
<code>/ayah/{surah}:{verse}</code>	Fetch verse text and page number	<code>/ayah/1:1</code>
<code>/ayah/{surah}:{verse}/{reader}</code>	Fetch audio URL for a verse	<code>/ayah/1:1/ar.alafasy</code>

6.2 Available Reciters:

#	Arabic Name	API Identifier
1	Mishary Alafasy	ar.alafasy
2	Abdul Basit	ar.abdulbasitmurattal
3	Abdurrahman Al-Sudais	ar.abdurrahmaansudais
4	Saood Ash-Shuraym	ar.saoodshuraym
5	Ali Al-Hudhaify	ar.hudhaify
6	Hani Rifai	ar.hanirifai
7	Maher Al Muaiqly	ar.mahermuaiqly

6.3 API Response Handling

All API calls use **HttpURLConnection** on background threads. The response is parsed using **JSONObject**. Key data extracted from responses:

- verse text (data.text) — displayed in the yellow verse card.
- page number (data.page) — used to load the correct Mushaf image from Assets.
- audio URL (data.audio) — passed to MediaPlayer for recitation playback.

Sample API Response (GET /ayah/1:1):

```
{
  "code": 200,
  "status": "OK",
  "data": {
    "number": 1,
    "text": "بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ",
    "surah": { "number": 1, "name": "سُورَةُ الْفَاتِحَةِ" },
    "numberInSurah": 1,
    "page": 1
  }
}
```

Extracted fields:

- data.text → displayed in the yellow verse card.
- data.page → matched with /assets/quran_pages_main/{page}.png.
- data.audio → passed to MediaPlayer (when using reciter endpoint).

6.4 Error Handling

When the API call fails (no internet, timeout, or non-200 response), the app shows a Toast message 'Error: No Internet' and logs the error. The app continues functioning with any previously cached data in the SQLite database.

7. Extra Feature: Quiz Mode

Quiz Mode is the Group 13 extra feature. After completing a memorization session and reaching the Completion screen, the user can tap 'Start Quiz' to test their memory.

7.1 How It Works

- The QuizActivity receives the Surah number, name, start verse, and end verse from the Intent.
- It queries the verses table in SQLite to fetch only the verses within the memorized range.
- If fewer than 4 verses are available, it shows a message and exits gracefully.
- 3 random questions are generated from the fetched verses.
- Each question shows the first 20 characters of a verse and asks the user to select the full verse.
- 4 options are displayed: 1 correct answer and 3 random wrong answers from the same range.
- Correct answers turn the button GREEN, wrong answers turn RED, and the correct answer is highlighted.
- After 3 questions, a dialog shows the final score with an encouraging message.
- The user can retry the quiz or return to the home screen.

7.2 Score Messages

Score	Message
3/3	Excellent! Expert memorizer!
2/3	Very Good! Keep going!
1/3	Review the verses and try again!
0/3	Review the verses carefully!

8. Technologies Used

Technology / Library	Purpose
Java	Primary programming language
Android SDK 34	Target Android version
SQLite	Local persistent storage
AlQuran Cloud API	External Quran data source
URLConnection	HTTP requests to the API
JSONObject	Parsing API responses
MediaPlayer	Audio playback of recitations
SharedPreferences	Storing settings (theme, font, reciter)
ViewPager2	Swipeable How-to-Use slides
RecyclerView	Surah list and slide adapters

Project Structure:

- 8 Activity classes + 1 BaseActivity for shared font scaling
- 1 DatabaseHelper class extending SQLiteOpenHelper
- 1 SurahAdapter + 1 SurahModel for the RecyclerView
- 14 XML layout files
- 604 Mushaf page images stored in assets/quran_pages_main/

9. Conclusion

Little Quran Stars successfully implements all three required project phases along with the Group 13 extra feature (Quiz Mode). The application provides a complete, child-friendly Quran memorization experience with the following technical achievements:

Component	Implementation
UI / UX	8 screens with consistent green theme, RTL support, night mode, font scaling
Core Logic	Verse range selection, repeat control, auto-play, navigation, progress tracking
SQLite	2 tables (verses + user_progress), 5 CRUD functions, per-Surah progress storage
API	AlQuran Cloud API for text, audio, and page number; 7 reciters supported
Extra Feature	Quiz Mode with 3 MCQ questions from memorized verse range